# ES32079 Financial Derivatives
# Problem Set 2 Solutions

## Question 1

Differentiate between forward, futures, and options contracts in terms of:

- Standardization
- Settlement
- Credit risk
- Typical market participants

### Solution

- **Standardization:**
  Forwards are private, customizable agreements between two parties. Futures are standardized contracts traded on organized exchanges. Options are also standardized when exchange-traded but can be customized in OTC markets.
- **Settlement:**
  Forwards are typically settled at maturity by physical delivery or cash settlement. Futures are marked to market daily, with gains and losses settled each day through a clearing house. For exchange-traded options, the clearinghouse substantially reduces counterparty risk. For OTC options, the buyer may suffer a loss if the option writer defaults. The buyer's maximum loss from movements in the underlying price is normally the premium paid.
- **Credit risk:**
  Forwards carry significant counterparty risk because they are private contracts without an intermediary. Futures have minimal credit risk since the exchange clearing house guarantees performance. Option buyers face no default risk beyond the premium paid, while writers have potential exposure unless collateralized.
- **Typical market participants:**
  Forwards are used mainly by corporations and financial institutions for tailored hedging needs. Futures attract hedgers and speculators requiring liquidity and transparency. Options are widely used by hedgers, speculators, and portfolio managers seeking asymmetric payoff structures.

# Question 2

A manufacturing company uses large quantities of copper in its production process and is concerned about rising copper prices over the next six months.

- What derivative instrument could the company use to hedge this price risk?
- Would it take a long or short position?
- Identify one potential risk that remains even after hedging.

## Solution

- The company could use a futures contract (or a forward contract) on copper to hedge against rising prices.
- The company should take a long position in copper futures, locking in the purchase price for future delivery. This protects it from potential price increases.
- The hedge may not be perfect due to basis risk, the futures price and the company's actual purchase price may not move identically. Other residual risks include differences in contract maturity, quantity, or quality relative to the firm's actual copper needs.

# Question 3

A non-dividend-paying stock is currently priced at \$50. The risk-free rate is 6% per annum, with continuous compounding.

- Calculate the 6-month forward price for the stock.
- If the time to maturity were instead 1 year, what would the forward price be?

## Solution

We use the forward pricing formula for a non-dividend-paying asset under continuous compounding:

$$F_0 = S_0 e^{rT}$$

where $S_0 = 50$, $r = 0.06$, and $T$ is the time to maturity (in years).

- **For 6 months ($T = 0.5$):**

$$F_0 = 50e^{0.06\times 0.5} = 50e^{0.03} = 50 \times 1.03045 = \mathbf{51.52}$$

- **For 1 year ($T = 1$):**

$$F_0 = 50e^{0.06\times 1} = 50e^{0.06} = 50 \times 1.06184 = \mathbf{53.09}$$

Thus, the 6-month forward price is approximately \$51.52 and the 1-year forward price is approximately \$53.09.

# Question 4

Explain briefly why the forward price may differ from the expected future spot price. Which forces ensure that the forward price is not arbitrageable?

## Solution

The **forward price** and the **expected future spot price** are different concepts.

The forward price is determined by the current spot price and the cost of carrying the asset until the contract matures. For a non-dividend-paying investment asset, the no-arbitrage forward price is

$$F_0 = S_0 e^{rT}.$$

Here, $S_0$ is the current spot price, $r$ is the continuously compounded risk-free rate, and $T$ is the time to maturity. The forward price is determined by the no-arbitrage relationship, whereas the expected future spot price reflects market expectations and risk premia.

By contrast, the expected future spot price, $\mathbb{E}[S_T]$, is the future spot price expected on average by market participants. It depends on expectations about future market conditions and may include a risk premium. Therefore, in general,

$$F_0 \neq \mathbb{E}[S_T].$$

The forward price is kept close to its theoretical value by arbitrage.
If

$$F_0 \neq S_0 e^{rT},$$

an arbitrage opportunity may exist.

- If the forward price is above its no-arbitrage value, arbitrageurs will tend to sell forward contracts and take an offsetting position in the underlying asset.
- If the forward price is below its no-arbitrage value, arbitrageurs will tend to buy forward contracts and take an offsetting position in the underlying asset.

This trading activity helps move the forward price back towards its no-arbitrage value.

# Question 5

The current futures price for a Treasury bond deliverable in 3 months is 101.50. The bond's coupon is 6% per annum, with semiannual payments, and the next coupon is due in 4 months. The risk-free rate is 5% (continuous compounding).

- Explain qualitatively what factors might cause the current futures price (101.50) to be higher or lower than the current spot price of the bond.

- How might the timing of the next coupon payment affect the relationship between the spot price and the futures price?
- If the bond were zero-coupon, would you expect the futures price to be higher or lower than the spot price? Briefly explain.

### Solution

The relationship between the spot price and the futures price depends mainly on the cost of carrying the bond and any coupon income received before the futures contract expires.

- If financing costs are greater than the coupon income received during the contract period, the futures price will tend to be higher than the spot price.
- If coupon income is greater than financing costs, the futures price may be lower than the spot price.

In this example, the futures contract expires in three months, but the next coupon is paid in four months. Therefore, no coupon is received during the life of the futures contract. With a positive interest rate, the financing cost is positive, so the futures price would normally be expected to be higher than the current spot price.

If the bond were a zero-coupon bond, there would also be no coupon income. Under the simplified no-arbitrage model, its futures price would therefore be

$$F_0 = S_0 e^{rT}.$$

Since $r > 0$, this implies

$$F_0 > S_0.$$

Thus, the futures price of a zero-coupon bond would normally be higher than its spot price when interest rates are positive.

## Question 6

Define the following terms clearly:

- Spot rate
- Forward rate
- Par yield
- Zero-coupon rate

Briefly describe how these rates differ from one another in interpretation or use.

## Solution

- **Spot rate:** The interest rate agreed today for a zero-coupon investment that starts today and matures at a specified future date. It represents the annualized return on a risk-free bond that pays a single amount at maturity. The spot rate is also called the zero rate or zero-coupon rate for that maturity.
- **Forward rate:** The implied future interest rate between two future dates, derived from spot rates. It is the future rate implied by today's zero curve under no-arbitrage. It is not necessarily a prediction of the future market rate.
- **Par yield:** The coupon rate at which a bond would trade exactly at par value (price = face value). It represents an average of spot rates weighted by the bond's cash flows.
- **Zero-coupon rate:** The rate of return on a zero-coupon (discount) bond, paying a single lump sum at maturity. It is synonymous with the spot rate for that maturity.

**Differences in interpretation:** Spot (zero) rates apply to investments from today to a given maturity, forward rates apply to periods *between* future dates, and par yields represent the coupon rate consistent with current spot rates for a bond priced at par.

# Question 7

You observe the following Treasury bond data:

| Maturity (years) | Coupon Rate (annual) | Price per 100 par |
|---|---|---|
| 0.5 | 0% | 97.00 |
| 1.0 | 6% | 101.50 |
| 1.5 | 8% | 104.00 |

Assume semiannually coupon payments.

- Compute the 0.5-year zero rate.
- Using that, estimate the 1-year zero rate.
- Briefly explain what the next step would be if we wanted to extend the zero curve to longer maturities.

## Solution

Assume that coupons are paid semiannually and that zero rates are continuously compounded.

Let $P(0,T)$ denote the discount factor for maturity $T$. The relationship between the discount factor and the continuously compounded zero rate is

$$P(0,T) = e^{-R(T)T}.$$

1. **The 0.5-year zero rate**

   The 0.5-year zero-coupon bond pays \$100 at maturity and has a price of \$97:

   $$97 = 100e^{-R(0.5)\times 0.5}.$$

   Therefore,

   $$R(0.5) = -\frac{\ln(97/100)}{0.5} = 0.0609.$$

   Thus,

   $$\boxed{R(0.5) \approx 6.09\%}.$$

2. **The one-year zero rate**

   The 6% bond pays coupons semiannually, so each coupon is

   $$\frac{6\%}{2} \times 100 = \$3.$$

   Its cash flows are \$3 after 0.5 years and \$103 after one year. Therefore,

   $$101.50 = 3e^{-0.0609\times 0.5} + 103e^{-R(1)\times 1}.$$

   Since

   $$e^{-0.0609\times 0.5} = 0.97,$$

   we have

   $$101.50 = 3 \times 0.97 + 103e^{-R(1)}.$$

   Therefore,

   $$e^{-R(1)} = \frac{101.50 - 2.91}{103} = 0.95718.$$

   Hence,

   $$R(1) = -\ln(0.95718) = 0.0438.$$

   Thus,

   $$\boxed{R(1) \approx 4.38\%}.$$

3. **Extending the zero curve**

   To find the 1.5-year zero rate, use the 1.5-year bond and the zero rates already calculated for 0.5 and 1 year.

   The 8% bond pays \$4 every six months, so its cash flows are \$4 at 0.5 years, \$4 at 1 year, and \$104 at 1.5 years.

   Its price equation is

$$104 = 4e^{-R(0.5)\times 0.5} + 4e^{-R(1)\times 1} + 104e^{-R(1.5)\times 1.5}.$$

   The first two zero rates are already known, so the equation can be solved for the 1.5-year zero rate. This step-by-step process is called **bootstrapping**.